

Assessment Tool 2: Scenario-Based Questions

1. Media Company Video Streaming Platform Migration

Understanding of the Scenario

The media company needs a cloud architecture that can support millions of concurrent users while minimizing buffering and ensuring high availability. The solution must also optimize costs during fluctuating demand.

Risks / Issues

- Heavy traffic during live events.
- Buffering due to server overload.
- High infrastructure costs.
- Network latency affecting user experience.

Cloud Architecture Design

- Use **Compute Instances** to host streaming servers.
- Store videos in **Cloud Object Storage**.
- Deploy a **Content Delivery Network (CDN)** to cache videos at edge locations.
- Place a **Load Balancer** in front of application servers.
- Configure **Auto Scaling Groups** to automatically add or remove servers based on CPU utilization or network traffic.

Application of Concepts

- **Load Balancing** distributes user requests across multiple servers, preventing overload.
- **Auto Scaling** automatically increases or decreases server instances based on demand.
- **Caching** through CDN stores frequently accessed videos near users, reducing latency and buffering.

Cost Optimization

- Use **Pay-as-you-Go** pricing.
- Enable Auto Scaling to avoid paying for idle servers.
- Store rarely accessed videos in low-cost archival storage.
- Monitor usage and remove unused resources.

Decision Justification

This architecture ensures high performance, scalability, fault tolerance, and cost efficiency while maintaining a smooth streaming experience.

Conclusion

A cloud architecture with CDN, load balancing, auto scaling, and cloud storage provides reliable, scalable, and cost-effective video streaming.

2.FinTech Startup Using Containers Across Multiple Clouds

Understanding of the Scenario

The startup requires consistent application deployment across different cloud providers while maintaining security and scalability.

Risks / Issues

- Inconsistent environments.
- Complex deployment management.
- Service communication issues.
- Security differences between cloud providers.

Proposed Solution

- Package applications using **Docker containers**.
- Use **Kubernetes** for container orchestration.
- Deploy Kubernetes clusters across multiple cloud providers.
- Use Infrastructure as Code for consistent deployments.

Application of Concepts

- Kubernetes automates deployment, scaling, and recovery.
- Service Discovery enables containers to communicate using DNS.
- Horizontal Pod Autoscaler increases or decreases containers automatically based on workload.

Benefits

- High availability.
- Better scalability.
- Vendor independence.
- Faster deployment.

Risks

- Increased management complexity.
- Network latency between clouds.
- Higher operational costs.

- Security policy synchronization challenges.

Decision Justification

Container orchestration provides consistent deployments, efficient scaling, and easier application management across multiple cloud providers.

Conclusion

Docker with Kubernetes enables secure, scalable, and portable multi-cloud application deployment.

3.E-Learning Platform Backup and Disaster Recovery

Understanding of the Scenario

The organization lost important data due to accidental deletion and requires an effective backup and disaster recovery strategy.

Risks / Issues

- Permanent data loss.
- Service downtime.
- Business interruption.
- Loss of student records.

Solution Design

- Enable automatic cloud backups.
- Replicate data across multiple regions.
- Enable object versioning.
- Schedule daily incremental and weekly full backups.

Application of Concepts

- **Cloud Backup** stores copies securely.
- **Replication** creates copies in different regions.
- **Versioning** allows recovery of deleted or modified files.
- Disaster Recovery automatically restores services after failures.

RTO and RPO

- **Recovery Time Objective (RTO):** Minimize downtime using automated recovery.
- **Recovery Point Objective (RPO):** Frequent backups ensure minimal data loss.

Business Continuity Steps

4. Perform regular backups.
5. Test disaster recovery plans.
6. Use multi-region replication.

4. Monitor backup status.
5. Automate recovery procedures.

Decision Justification

Backup, replication, and versioning ensure quick recovery with minimal data loss.

Conclusion

Cloud-native backup and disaster recovery provide reliable protection and ensure continuous educational services.

6. Government Hybrid Cloud Architecture

Understanding of the Scenario

Sensitive government applications must remain on-premises, while non-sensitive services can be deployed in the public cloud.

Risks / Issues

- Data leakage.
- Compliance violations.
- Unauthorized access.
- Secure communication challenges.

Hybrid Cloud Design

- Store confidential information in the private cloud/on-premises.
- Host public services in the public cloud.
- Connect environments using secure VPN or dedicated private connections.

Application of Concepts

- Implement **Identity and Access Management (IAM)**.
- Enable **Single Sign-On (SSO)**.
- Apply **Role-Based Access Control (RBAC)**.
- Encrypt data during storage and transmission using strong encryption standards.

Challenges

- Managing multiple environments.
- Policy synchronization.
- Increased operational complexity.
- Compliance monitoring.

Decision Justification

Hybrid cloud balances security, compliance, and scalability while maintaining operational flexibility.

Conclusion

A secure hybrid cloud architecture protects sensitive information while utilizing the scalability of the public cloud.

5.SaaS Multi-Region High Availability

Understanding of the Scenario

The SaaS provider requires continuous application availability even if an entire cloud region fails.

Risks / Issues

- Regional outages.
- Data loss.
- Service downtime.
- Customer dissatisfaction.

Multi-Region Deployment Strategy

- Deploy application servers in multiple cloud regions.
- Use global load balancing.
- Configure automatic failover.
- Store replicated databases across regions.

Application of Concepts

- **Data Replication** keeps identical copies in multiple regions.
- **Failover** automatically redirects users to healthy regions.
- **Load Balancing** distributes traffic globally.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network.
- Multi-region deployment ensures fault tolerance, disaster and uninterrupted service recovery availability.

- Configure automatic alerts for failures.

Conclusion

A multi-region architecture with replication, failover, monitoring, and load balancing provides maximum availability and reliability for SaaS applications.

- Perform health checks continuously.

Decision Justification